Lab Report on Association Rule

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1 Objective

The objective of association rule mining is to discover relationships and co-occurrence patterns among products in transaction data. It identifies products that tend to appear together in the same transaction and represents these relationships as rules of the form $A \rightarrow B$.

The strength of association rules is evaluated using three important metrics: support, confidence, and lift. These metrics help retailers understand purchasing patterns and make decisions about product placement, cross-selling, product bundling, and promotional strategies.

In this experiment, the FP-Growth algorithm was applied to the Retail Transactions Dataset to identify frequent itemsets and generate association rules at different minimum support thresholds.

2 Dataset

The dataset used in this experiment is the Retail Transactions Dataset published on Kaggle. The dataset contains 1,000,000 transaction records and 13 columns.

The columns are:

- Transaction_ID
- Date
- Customer_Name
- Product
- Total_Items
- Total_Cost
- Payment_Method

- City
- Store_Type
- Discount_Applied
- Customer_Category
- Season
- Promotion

The `Product` column contains lists of products purchased in each transaction. For example, a transaction may contain Ketchup, Shaving Cream, and Light Bulbs.

Each transaction was treated as a basket of products. The product information was used to discover purchasing patterns, while the other columns were not directly used in the association rule mining process.

The original dataset contains 1,000,000 transactions. The analysis identified 81 unique products in the transaction data used to generate the reported results.

3 Preprocessing

The `Product` column was used for preprocessing. Since product lists were stored as text, each list was converted into a Python list using `ast.literal_eval`. Duplicate products within the same transaction were removed to ensure that each product was represented only once per basket.

Next, the transaction lists were converted into a Boolean matrix using one-hot encoding with the `TransactionEncoder` class from the `mlxtend` library.

In the resulting matrix, each row represents a transaction and each column represents a unique product. A value of `True` indicates that the product appears in the transaction, whereas `False` indicates that the product is absent.

This Boolean representation is suitable for frequent itemset mining because it allows the algorithm to calculate the proportion of transactions containing particular products or product combinations.

The analysis identified 81 unique product columns. The FP-Growth algorithm was then applied to the preprocessed transaction data.

4 Frequent Itemsets

The FP-Growth algorithm was used to discover frequent itemsets at four minimum support thresholds: 0.01, 0.005, 0.002, and 0.001.

The results are presented in Table 1.

Table 1: Frequent itemsets at different support thresholds.

Minimum Support	1-Itemsets	2-Itemsets	3-Itemsets	Total
0.010	81	0	0	81
0.005	81	0	0	81
0.002	81	79	0	160
0.001	81	3062	0	3143

At minimum support thresholds of 0.01 and 0.005, only single-item frequent itemsets were identified. At 0.002, the algorithm identified 79 frequent two-item combinations.

When the threshold was reduced to 0.001, the number of frequent two-item combinations increased to 3,062. No three-item frequent itemsets were reported at the tested thresholds.

Table 2 presents five frequent two-item combinations from the experiment.

Table 2: Top frequent two-item combinations.

Frequent Itemset	Support
Canned Soup, Toothpaste	0.00261
Peanut Butter, Toothpaste	0.00259
Toothpaste, BBQ Sauce	0.00259
Toothpaste, Ironing Board	0.00255
Toothpaste, Cleaning Spray	0.00253

These results demonstrate that reducing the minimum support threshold allows more combinations of products to qualify as frequent itemsets. However, a lower threshold can also introduce combinations that occur in relatively few transactions.

5 Association Rules

Association rules were generated from the frequent itemsets using a minimum support threshold of 0.001. The reported results contained 6,124 association rules.

The maximum lift was 1.166, while the mean lift was 0.903. Three representative rules are presented in Table 3.

Table 3: Representative association rules.

Antecedent	Consequent	Support	Confidence	Lift
Ironing Board	Vacuum Cleaner	0.00147	0.041119	1.165832
Vacuum Cleaner	Ironing Board	0.00147	0.041678	1.165832
Lawn Mower	Shower Gel	0.00147	0.041667	1.152924

5.1 Support

Support measures the proportion of all transactions containing both the antecedent and the consequent. For example, a support value of 0.00147 corresponds to approximately 0.147% of transactions containing both products.

5.2 Confidence

Confidence measures the conditional probability of observing the consequent when the antecedent is present.

For the rule

$$IroningBoard \rightarrow VacuumCleaner,$$

the confidence is 0.041119, or approximately 4.11%. This means that approximately 4.11% of transactions containing an Ironing Board also contain a Vacuum Cleaner.

5.3 Lift

Lift compares the confidence of an association rule with the overall probability of observing its consequent.

$$Lift(A \rightarrow B) = \frac{P(B \mid A)}{P(B)}$$

A lift greater than 1 indicates a positive association, a lift equal to 1 indicates statistical independence under the measured relationship, and a lift below 1 indicates a negative association.

The first rule has a lift of 1.165832. Therefore, the conditional probability of purchasing a Vacuum Cleaner when an Ironing Board is present is approximately 1.166 times the baseline probability of purchasing a Vacuum Cleaner.

6 Discussion

The results show that the dataset contains some positive associations between products. However, the observed relationships are relatively weak. The maximum lift of 1.166 is only moderately above 1, and the mean lift of 0.903 indicates that the average reported rule does not show positive association.

From a business perspective, a retailer could investigate the Ironing Board and Vacuum Cleaner combination for cross-selling recommendations or product placement. Nevertheless, the low support and confidence values indicate that these products occur together infrequently and that the conditional probability remains low.

The results also show the importance of selecting an appropriate minimum support threshold. At 0.005, the analysis found 81 frequent itemsets, whereas at 0.001, the total increased to 3,143. Lowering the threshold revealed many additional two-item combinations.

Despite the larger number of itemsets, more patterns do not necessarily mean more useful business insights. Rare combinations may be unstable or may occur by chance.

6.1 Limitations

The experiment has several limitations:

1. **Weak associations:** The maximum lift of 1.166 suggests that the strongest observed relationship is relatively modest.
2. **Low support and confidence:** The representative rules have low support and confidence, limiting their immediate business value.
3. **No causal interpretation:** Association rules identify co-occurrence patterns but cannot establish that purchasing one product causes the purchase of another.
4. **Threshold sensitivity:** The number of frequent itemsets increases substantially as the minimum support threshold decreases. Some resulting combinations may be rare or uninformative.
5. **Limited itemset size:** No three-item frequent itemsets were reported at the tested thresholds, so larger product combinations could not be discussed using these results.
6. **Limited context:** The analysis focused on product co-occurrence and did not directly investigate profitability, product quantities, customer characteristics, or seasonal purchasing patterns.

Consequently, retailers should evaluate support, confidence, lift, and practical relevance together before using association rules to make business decisions.

7 Conclusion

This experiment demonstrated the application of association rule mining to retail transaction data. The product lists were converted into a Boolean matrix, and the FP-Growth algorithm was used to identify frequent itemsets at four minimum support thresholds.

The total number of frequent itemsets increased from 81 at minimum support 0.01 to 3,143 at minimum support 0.001. The reported analysis generated 6,124 association rules, with a maximum lift of 1.166 and a mean lift of 0.903.

The findings demonstrate that lowering the minimum support threshold allows the discovery of more product combinations. However, many of the observed associations are weak and may have limited business value.

Overall, the experiment provided practical experience in transaction preprocessing, Boolean encoding, frequent itemset generation, and the interpretation of support, confidence, and lift. Further analysis using validated transaction data and additional business context could help determine whether these patterns are useful for real-world retail decisions.

Dataset Link

Retail Transactions Dataset – Kaggle:

https://www.kaggle.com/datasets/prasad22/retail-transactions-dataset